

Dee Aein

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EDUCATION

San Jose State University

Aug 2025 - Oct 2027

Bachelor of Science, Computer Science (GPA: 3.93)

San Jose, CA

- **Coursework:** Data Structures & Algorithms, Database Systems, Linear Algebra, Graph Theory, Object Oriented Programming, Advanced Python, Artificial Intelligence (search logic, Bayesian reasoning, reinforcement learning), Machine Learning (regression, classification, MLP, CNN, autoencoders, attention mechanism, model evaluation)

SKILLS

- **Languages:** Java, Python, SQL
- **Technologies:** REST APIs, FastAPI, Flask, PostgreSQL, MySQL, SQLAlchemy, Alembic, Docker, AWS (ECS/Fargate, RDS, S3, CloudFront, ECR, ALB, IAM), RAG, Sentence Transformers, OpenAI API, scikit-learn, NumPy, pandas, Matplotlib, JavaFX, Java Servlets, Git, GitHub, Maven

PROJECT EXPERIENCE

AI Codebase Assistant (RAG)

Jul 2026 – Present

Python, FastAPI, PostgreSQL, pgvector, OpenAI API, Sentence Transformers, SQLAlchemy, Pydantic, Docker, AWS ECS/Fargate, RDS, S3, CloudFront, Pydantic

- Engineered an end-to-end repository ingestion pipeline that processed 55 source files into 158 searchable code chunks in 27 seconds by integrating the GitHub API, file filtering, AST-based chunking, embeddings, and PostgreSQL storage.
- Achieved 100% top-10 and 86.7% top-2 retrieval success across 15 repository-specific questions on 2 codebases by implementing semantic retrieval with Sentence Transformer embeddings and pgvector cosine-similarity search.
- Achieved 100% grounded responses across 15 manually evaluated answers by constraining OpenAI-generated responses to retrieved code context and returning exact file- and line-level citations through REST API endpoints.
- Deployed a reproducible full-stack application to AWS by containerizing services with Docker and integrating ECR, ECS/Fargate, RDS PostgreSQL, S3, CloudFront, an Application Load Balancer, IAM, Secrets Manager, and security-group-based networking.

MedIMS – 3-Tier Inventory Management System

Mar 2026 - Apr 2026

Java, Java Servlets, MySQL, JDBC, BCrypt, AWS

- Led a 3-person team to deliver a role-based hospital inventory system on schedule by coordinating development workflows, delegating tasks, and driving project execution
- Designed a normalized MySQL data layer across 15 relational tables by translating ER diagrams into DAO-based persistence logic with complex JOIN queries and prepared statements
- Built backend services across 11 REST endpoints by implementing Java Servlet for CRUD operations, submission/approval workflows, session-based authentication, and BCrypt password hashing.

Snake Game – Real-Time Interactive Application

Oct 2025 - Nov 2025

Java, JavaFx, OOP, Event-Driven Programming

- Engineered a real-time JavaFx game sustaining 50 FPS by implementing an AnimationTimer game loop, collision detection, and 20-pixel grid-based state management
- Improved movement reliability and state consistency by modeling snake segments with custom SnakeBody objects and dynamic list updates, preventing invalid reverse-direction inputs and maintaining synchronized navigation and scoring state.

Rental Marketplace System – CLI-Based Backend Simulation

Aug 2025 - Sep 2025

Java, OOP, HashMap, Exception Handling

- Led a 5-person team to build a role-based rental platform supporting 9 vendor and customer operations by coordinating development and implementing inventory, pricing, rental, and return workflows over 150+ in-memory product instances.
- Prevented identifier collisions and enforced consistent business rules across by designing a modular object-oriented architecture with an abstract Product class, Rentable/Buyable interfaces, composite-key mapping and exception-based validations.

WORK EXPERIENCE

Math, Science & Technology Tutor

Sep 2023 - Jan 2025

De Anza college

- Led weekly tutoring support for 30+ students across math and technical subjects, applying data structures, algorithms, and structured reasoning to strengthen problem-solving methods to strengthen analytical skills
- Mentored 2–3 students per quarter over 1.5 years through recurring 1:1 sessions by evaluating progress, adapting instruction, and guiding students toward stronger performance in STEM coursework

ICC Representative, Competitive Programming Club

Mar 2024 - Aug 2025

De Anza College

- Represented the Competitive Programming club in Inter-Club Council meetings over 1.5 years, coordinating with campus organizations and administrators to communicate club initiatives and operational requirements
- Led weekly algorithm problem-solving sessions for ~10 students, guiding participants from brute-force solutions to optimized data-structure and algorithm approaches while strengthening time-complexity analysis.